Course Code	**DESIGN OF STEEL STRUCTURES**	L	T	P	C
23A01601		3	0	0	3

Course Objectives:

The objectives of this course are to make the student to:

1. **Understand** the properties, types, and applications of structural steelin construction.
2. **Analyze** the behavior and design of bolted and welded connections for steel structures.
3. **Design** tension and compression members, including built-up members and column bases.
4. **Develop** steel structural elements such as beams, plate girders, roof trusses, and gantry girders.
5. **Apply** plastic analysis concepts to the design of continuous beams and portal frames.

Course Outcomes:

Upon successful completion of this course, students will be able to:

1. **Explain** the properties of structural steel, types of sections, and the concept of limit state design.
2. **Analyze and design** bolted and welded connections for structural steel members.
3. **Design** tension and compression members, including built-up sections and column bases.
4. **Develop** design solutions for beams, plate girders, roof trusses, and gantry girders.
5. **Perform** plastic analysis and design of continuous beams and portal frames.

CO – PO Articulation Matrix

Course Outcomes	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO1**	**PSO2**
CO -1	**3**	**2**	**2**	**2**	**1**	**1**	**1**	**-**	**1**	**-**	**2**	**2**	**2**	**2**
CO -2	**3**	**2**	**2**	**2**	**1**	**1**	**1**	**-**	**1**	**-**	**2**	**2**	**1**	**1**
CO -3	**3**	**2**	**2**	**2**	**1**	**-**	**-**	**-**	**1**	**-**	**2**	**2**	**2**	**1**
CO -4	**3**	**2**	**2**	**2**	**1**	**-**	**-**	**-**	**1**	**-**	**2**	**2**	**1**	**1**
CO -5	**3**	**2**	**2**	**2**	**1**	**-**	**-**	**-**	**1**	**-**	**2**	**2**	**2**	**2**

UNIT – I

INTRODUCTION to STRUCTURAL STEEL and DESIGN of CONNECTIONS

General -Types ofSteel -Properties of Structural Steel - I.S. Rolled Sections - Concept of Limit State Design - Design of Simple and Eccentric Bolted and Welded Connections - Types of Failure and Efficiency of Joint – Prying Action - Introduction to HSFG bolts

UNIT – II

DESIGN of TENSION and COMPRESSION MEMBERS

Behaviour and Design of Simple and Built-Up Members Subjected to Tension - Shear Lag Effect Design of Lug Angles - Tension Splice - Behaviour of Short and Long Columns - Euler's Column Theory Design of Simple and Built-Up Compression Members With Lacings and Battens - Design of Column Bases - Slab Base and Gusseted Base

UNIT – III

DESIGN of BEAMS

Design of Laterally Supported and Unsupported Beams - Design of Built-Up Beams - Design of Plate Girders

UNIT – IV		
INDUSTRIAL STRUCTURES Design of Roof Trusses – Loads On Trusses – Purlin Design Using Angle and Channel Sections – Truss Design, Design of Joints and End Bearings–Design of Gantry Girder - Introduction to Pre-Engineered Buildings		
UNIT – V		
PLASTIC ANALYSIS and DESIGN Introduction to Plastic Analysis - Theory of Plastic Analysis - Design of Continuous Beams and Portal Frames Using Plastic Design Approach.		
TEXT BOOKS:		
1. Duggal S.K., Design of Steel Structures, Tata McGraw Hill, Publishing Co. Ltd., New Delhi, 2010 2. Bhavikatti S.S, Design of Steel Structures, Ik International Publishing House, New Delhi, 2017.		
REFRENCE BOOKS:		
1. Gambhir M L, Fundamentals of Structural Steel Design, McGraw Hill Education India Pvt Limited, 2013 2. Jack C. McCormac& Stephen F. Csernak - Structural Steel Design, Pearson, 7th Edition, 2023. 3. William T. Segui&FaridSoleimani - Steel Design, Cengage, 7th Edition, 2023. 4. SarwarAlamRaz, Structural Design in Steel, New Age International Publishers, 2014 5. Subramanian N, Design of Steel Structures, Oxford University Press, New Delhi, 2016		
Online Learning Resources:		
https://nptel.ac.in/courses/105105162		